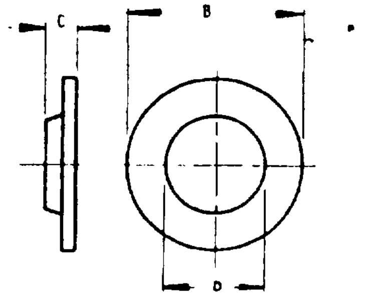
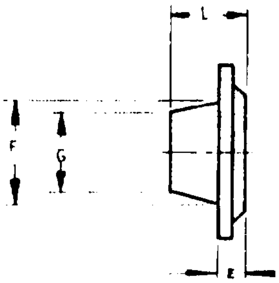
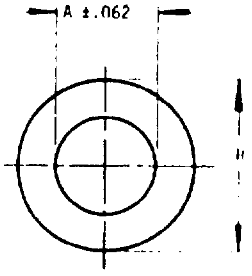
| FED. SUP CLASS |
|---|
| 5325 |



PLAIN WASHER

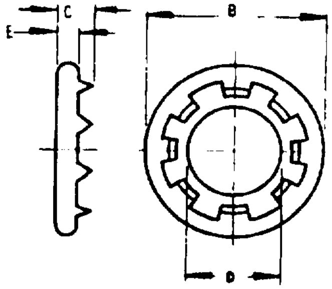


PLAIN GROMMET

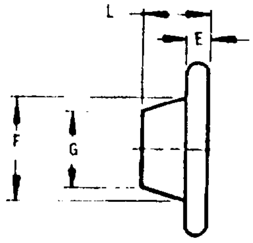


PLAIN GROMMET

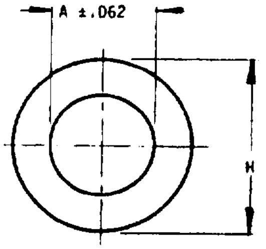
| ASSY DASH NO. | | | DETAILED PART NO. | | | | | | TRADE SIZE NO. | A DIA AFTER INSERTION | B ±.005 DIA | C ±.005 | D ±.005 DIA | E ±.005 | F ±.005 OD | G ±.005 ID | H ±.005 OD | L ±.005 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AL | BRASS | | GROMMET | | | WASHER | | | | | | | | | | | | |
| | PLAIN | PLATED | AL | BRASS PLAIN | BRASS PLATED | AL | BRASS PLAIN | BRASS PLATED | | | | | | | | | | |
| A20 | B20 | BP20 | GA20 | GB20 | GBP20 | WA20 | WB20 | WBP20 | 00 | .156 | .460 | .029 | .218 | .035 | .218 | .176 | .425 | .170 |
| A10 | B10 | BP10 | GA10 | GB10 | GBP10 | WA10 | WB10 | WBP10 | 0 | .250 | .546 | .030 | .286 | | .290 | .240 | .545 | .210 |
| A1 | B1 | BP1 | GA1 | GB1 | GBP1 | WA1 | WB1 | WBP1 | 1 | .312 | .700 | .041 | .359 | | .375 | .286 | .687 | .220 |
| A2 | B2 | BP2 | GA2 | GB2 | GBP2 | WA2 | WB2 | WBP2 | 2 | .375 | .835 | .045 | .453 | | .475 | .362 | .815 | .250 |
| A3 | B3 | BP3 | GA3 | GB3 | GBP3 | WA3 | WB3 | WBP3 | 3 | .438 | .955 | .050 | .530 | .050 | .545 | .433 | .970 | .345 |
| A4 | B4 | BP4 | GA4 | GB4 | GBP4 | WA4 | WB4 | WBP4 | 4 | .500 | 1.062 | | .585 | .046 | .630 | .486 | 1.050 | .420 |
| A5 | B5 | BP5 | GA5 | GB5 | GBP5 | WA5 | WB5 | WBP5 | 5 | .625 | 1.205 | .055 | .740 | .045 | .785 | .594 | 1.220 | .375 |



SPUR WASHER



SPUR GROMMET



ROLLED RIM GROMMET

| ASSY DASH NO. | | DETAILED PART NO. | | | | TRADE SIZE NO. | A DIA AFTER INSERTION | B ±.005 OD | C ±.005 | D ±.005 ID | E ±.005 | F ±.005 OD | G ±.005 ID | H ±.005 OD | L ±.020 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BRASS | | GROMMET BRASS | | WASHER BRASS | | | | | | | | | | | |
| PLAIN | PLATED | PLAIN | PLATED | PLAIN | PLATED | | | | | | | | | | |
| BS10 | BPS10 | GBS10 | GBPS10 | WBS10 | WBPS10 | 0 | .250 | .645 | .150 | .345 | .060 | .375 | .270 | .670 | .235 |
| BS1 | BPS1 | GBS1 | GBPS1 | WBS1 | WBPS1 | 1 | .312 | .790 | | .455 | .065 | .480 | .380 | .790 | .275 |
| BS2 | BPS2 | GBS2 | GBPS2 | WBS2 | WBPS2 | 2 | .375 | .890 | .156 | .512 | .070 | .532 | .425 | .890 | .315 |
| BS3 | BPS3 | GBS3 | GBPS3 | WBS3 | WBPS3 | 3 | .438 | .970 | .190 | .537 | .075 | .545 | .430 | .970 | .330 |
| BS4 | BPS4 | GBS4 | GBPS4 | WBS4 | WBPS4 | 4 | .500 | 1.120 | .205 | .645 | .080 | .695 | .560 | 1.120 | .405 |
| BS5 | BPS5 | GBS5 | GBPS5 | WBS5 | WBPS5 | 5 | .625 | 1.270 | .225 | .777 | .085 | .800 | .600 | 1.285 | .455 |

MATERIAL: BRASS OR ALUMINUM. SEE PROCUREMENT SPECIFICATION.

FINISH: ALUMINUM, ANODIZE IN ACCORDANCE WITH MIL-A-8625; BRASS, CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2 ONLY WHEN SPECIFIED.

EXAMPLES OF PART NUMBERS: MS20230A20 - NO. 00 SIZE, PLAIN GROMMET ASSEMBLY, ALUMINUM.
MS20230BPS10 - NO. 0 SIZE, SPUR GROMMET ASSEMBLY, BRASS, CADMIUM PLATED.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010. Ⓓ ~~CANCELED~~ ~~AFTER 14 OCTOBER 1966. USE MIL-G-16491.~~

THE ASSEMBLY SHALL CONSIST OF ONE GROMMET AND ONE WASHER OF THE SAME MATERIAL.

INTERCHANGEABILITY RELATION WITH AN230 GROMMETS: MS20230 GROMMETS AND AN230 GROMMETS OF LIKE DASH NUMBERS ARE UNIVERSALLY, FUNCTIONALLY, AND DIMENSIONALLY INTERCHANGEABLE.

Ⓓ **REINSTATE** AFTER 28 MARCH 1969

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

APPROVED 19 AUG 57

REVISED Ⓐ 7 OCT 59 Ⓑ 1 JAN 64 Ⓒ 16 OCT 66 Ⓓ 28 MAR 69

| | | |
|---|---|---|
| **P.A.**<br>**NAVY — AS**<br>**Other Cust**<br>USAF - 82 | **TITLE**<br>GROMMETS, METALLIC, PLAIN AND<br>SPUR, WITH WASHERS, TYPE I AND TYPE III | **MILITARY STANDARD**<br>**MS20230(ASG)** |
| **PROCUREMENT SPECIFICATION**<br>MIL-G-16491 | **SUPERSEDES:**<br>AN230 | **SHEET** 1 **OF** 1 |

USER SYMBOLS:

REVIEWER SYMBOLS:
NAVY-AS
USAF-82

"Review/user information is current as of the date of this document. For future coordination of changes to this document draft circulation should be based on the information in the current DODISS."

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AMSC N/A

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**DD** FORM 1 MAR 72 **672-1** (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 20000

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